EXPERIMENT 8: NMAP LIVE HOST DISCOVERY

AIM

To discover live hosts using Nmap scans based on ARP, ICMP, and TCP/UDP probes.

INTRODUCTION

Nmap (Network Mapper) is an open-source tool used to identify active hosts. Host discovery is performed before port scanning to identify systems that are online.

PROCEDURE

The following commands demonstrate different host-discovery methods.

1. ARP / LOCAL NETWORK DISCOVERY

```
$ sudo nmap -sn -PR 192.168.1.0/24
Nmap scan report for 192.168.1.1
Host is up (0.0030s latency).
Nmap scan report for 192.168.1.10
Host is up (0.0050s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned
```

ARP discovery is useful on a local Ethernet network because it checks which local devices respond to ARP requests.

2. ICMP ECHO DISCOVERY

```
$ nmap -sn -PE 192.168.1.0/24
Nmap scan report for 192.168.1.1
Host is up (0.0030s latency).
Nmap done: 256 IP addresses (1 host up) scanned
```

ICMP echo probes are similar to ping requests. A host may not respond if ICMP traffic is blocked.

3. TCP SYN DISCOVERY

```
$ nmap -sn -PS443 192.168.1.0/24
Nmap scan report for 192.168.1.10
Host is up (0.0040s latency).
Nmap done: 256 IP addresses (1 host up) scanned
```

This sends TCP SYN discovery probes to port 443. A TCP response can indicate that the target is reachable.

4. UDP DISCOVERY

```
$ sudo nmap -sn -PU 192.168.1.0/24
Nmap scan report for 192.168.1.1
Host is up (0.0060s latency).
Nmap done: 256 IP addresses (1 host up) scanned
```

UDP discovery sends UDP probes. Responses, including certain ICMP messages, can help Nmap determine whether a host is active.

OBSERVATIONS

- ARP, ICMP, TCP, and UDP probes use different methods to discover active hosts.
- Results depend on the network, host configuration, and firewall rules.
- The outputs shown are sample results; actual IP addresses and host counts may vary.

RESULT

Nmap live-host discovery was studied using ARP, ICMP, TCP SYN, and UDP probes, and the outputs were interpreted to understand how active hosts are identified.